



BRICKSPLAZA

EVERY BRICK, EVERY BUILD.

E-COMMERCE PLATFORM

TECHNICAL & PRODUCT SPECIFICATION

Version 1.0

A Specialised Operating Division of Besbpo Group

Next.js · NestJS · Python/FastAPI · Hosted on Render

01 — Executive Summary

This specification defines the full build for Bricksplaza's e-commerce platform: a headless, API-first storefront and back-office system serving 2,044 SKUs across 21 categories, with a customer-facing store, a dedicated Business Desk (admin back-office), and the supporting services (search, recommendations, document generation) needed to run it as a real trading platform — not a brochure site.

The brief called for comprehensive coverage across authentication, page inventory, payments, delivery, discovery, and agentic build process. This document delivers that: every section below is written at build-ready depth for a human-in-the-loop (HITL) agentic development process, with the detailed schemas, exact copy and pixel-level design left to the build phase itself, consistent with how Roofsteel and Bellwether SWE Plumbers were built.

Scale note: at 2,044 SKUs, 5 OAuth providers, 15 payment methods, a full CMS and Business Desk, and dedicated recommendation/search services, this platform is substantially larger in scope than Roofsteel's e-commerce build (which shipped at a real 35,521 LoC). Section 16 gives a like-for-like estimate.

02 — Platform Overview & Objectives

- Single-vendor, multi-tenant-ready storefront (Bricksplaza today; architecture allows a second Besbpo Group storefront to reuse the same platform later, as Roofsteel's stack was designed to)
- Serve Retail, Trade and Volume customer tiers from one catalogue, with tier-aware pricing already defined in the Pricing Engine (v1.0)
- Support both transactional checkout (add-to-cart → pay → deliver) and RFQ/bulk-quote workflows for large civil/industrial orders
- Launch serving Gauteng and KwaZulu-Natal with physical yard fulfilment, online-first nationwide across all 7 remaining provinces
- Built for a human-in-the-loop agentic development process: AI agents scaffold and implement against this spec; humans review, approve and merge every change (Section 20)

03 — Technology Stack

Application Layer

- Frontend (storefront): Next.js 14 (App Router), TypeScript, React, Tailwind CSS — SSR/ISR for SEO-critical pages (category, product, blog)
- Frontend (Business Desk / admin): a separate Next.js app sharing the same design system package, role-gated
- API layer: NestJS (TypeScript) as the API Gateway / BFF — REST for the storefront, GraphQL for the admin data-grid-heavy screens
- Microservices: Python/FastAPI for recommendations, search indexing, and PDF/document generation — the workloads best served by Python's ML/data ecosystem, mirroring the Next.js+NestJS+FastAPI pattern already proven on Roofsteel
- Background jobs: BullMQ (Node) workers for email, webhooks, carrier sync, and search re-indexing

Data & Infrastructure

- PostgreSQL via Supabase — primary transactional datastore (consistent with Group-wide infrastructure standard)
- Redis via Upstash — cache, session store, and BullMQ queue backend
- Typesense (self-hosted on Render) or Meilisearch — product search index; open-source, cost-effective, and fast enough for a 2,044-SKU catalogue without Algolia's per-search billing
- S3-compatible object storage (e.g. Cloudflare R2 or AWS S3 af-south-1) — product images, spec-sheet PDFs, invoices
- Apache Kafka (AWS MSK) — event bus for order/inventory/pricing events feeding analytics and the Business Desk in near-real-time, consistent with Group standard
- Hosting: Render (web services, background workers, cron jobs) — chosen per brief; Terraform/Terragrunt for IaC, ArgoCD-style GitOps for deploys where Render's native pipelines are insufficient

Third-Party Services

- Auth.js (NextAuth) — handles all 6 authentication methods (Section 5) from one library
- Resend or Postmark — transactional email (order confirmations, password resets, shipping updates)
- Twilio or a local SA aggregator (e.g. Clickatell) — SMS delivery notifications
- Sentry — error monitoring across all services
- PostHog or Plausible — product analytics (self-hostable, POPIA-friendlier than GA4)

04 — System Architecture

The platform is composed of three client surfaces (storefront, admin, and a shared mobile-responsive PWA build) talking to one NestJS API gateway, which fans out to domain services and three Python/FastAPI microservices for specialised workloads. All synchronous services are stateless and horizontally scalable on Render; state lives in PostgreSQL, Redis and the search index.

System Architecture — High Level

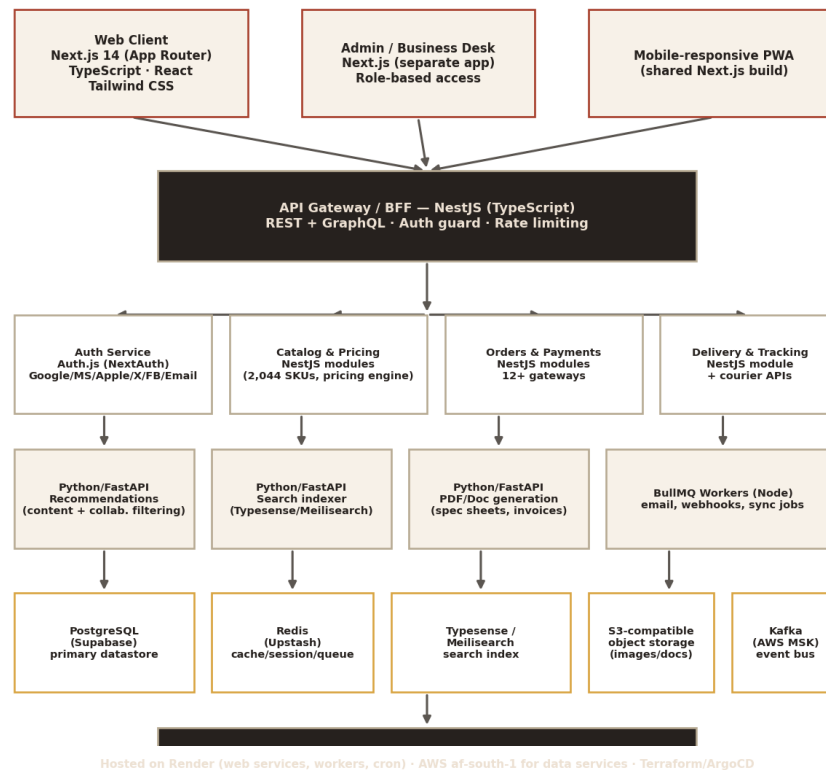


Figure 1 — High-level system architecture

05 — Authentication & Account System

Account creation and sign-in support five OAuth providers plus email/password, unified through Auth.js so the NestJS API only ever deals with one verified JWT regardless of provider.

Provider	Integration	Why it's included
Google	Google Identity Services (OAuth 2.0 / OIDC)	Most-used consumer login in SA; one-tap on Android

Provider	Integration	Why it's included
Microsoft	Microsoft Entra ID (Azure AD) via MSAL	Covers corporate/procurement buyers using Microsoft 365 accounts
Apple	Sign in with Apple (mandatory alongside other social logins on iOS per App Store guidelines)	Required if a native iOS app ships later; also a clean privacy-forward option
X (Twitter)	X OAuth 2.0 (API v2)	Secondary social login option
Facebook	Meta Login (Facebook Login SDK)	Still widely used across SA consumer demographics
Email / Password	Argon2id password hashing, email verification, rate-limited login	Baseline method; always available regardless of social outages

Authentication Flow

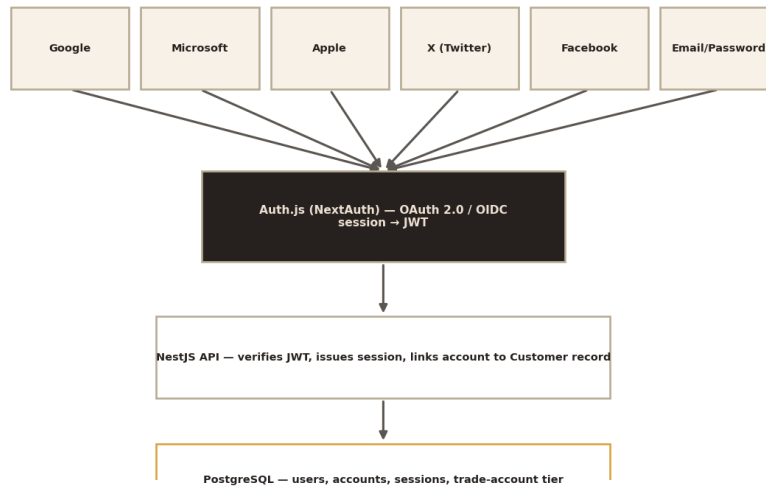


Figure 2 — Authentication flow, all providers converge on one session model

Account & Recovery Flows

- Sign up: social one-click, or email + password with mandatory email verification before first checkout
- Sign in: social, email/password, or a passwordless magic-link fallback for support-desk-assisted recovery
- Forgot password: time-limited token (15 min) emailed, rate-limited to prevent enumeration
- Account recovery: secondary flow for social-login-only accounts that lose provider access — verified via email + order history match, escalating to manual support review for trade accounts above a credit-limit threshold
- Session security: short-lived JWT + refresh token rotation; forced re-auth for payment-method changes and trade-account credit-limit changes
- Account merging: if a customer signs up with email first and later uses Google with the same address, accounts are merged with an explicit confirmation step, not silently

06 — Sitemap & Page Inventory

The full inventory below covers every page category requested — legal, accessibility, structural, account, payment, support, marketing, SEO, CMS editor screens, and the Business Desk back-office. Counts are for distinct templates; product and category pages then multiply out across the real catalogue (2,044 product pages, 21 category pages).

Category	Page Count	Includes
Legal	9	Terms & Conditions, Privacy Policy, Cookie Policy, Returns & Refunds Policy, Warranty Policy, POPIA Data Subject Rights, Acceptable Use Policy, Trade Account Terms, Sale of Goods Terms
Accessibility	1	Accessibility Statement (WCAG 2.1 AA target, Section 6 detail)
Structural / Core	11	Homepage, Category Listing (template × 21), Sub-category filter view, Product Detail (template × 2,044), Search Results, Cart, Checkout (multi-step), Order Confirmation, 404, 500, HTML Sitemap
Account	14	Sign Up, Sign In, Forgot Password, Reset Password, Account Recovery, Dashboard, Order History, Order Detail/Tracking, Saved Addresses, Saved Payment Methods, Wishlist, Trade Account Application, Notification Preferences, Profile Settings
Payment	3	Payment Confirmation, Payment Failed/Retry, Invoice/Receipt Download
Support	7	Help Centre/FAQ, Contact Us, Track My Order (public), Returns Request Form, Warranty Claim Form, Bulk Quote/RFQ Request, Live Chat widget (overlay, not a page)
Marketing	5	Promotion Landing Page template, Blog/Content Hub, Blog Post template, Project Case Studies/Gallery, Referral Program
SEO	3	Category SEO Landing template, Province/Location Landing template, XML Sitemap (machine-readable)
CMS (editor screens)	5	Content Editor, Media Library, Blog Post Editor, Promotion/Banner Manager, SEO Metadata Manager
Business Desk (admin)	13	Dashboard, Order Management, Inventory Management, Product/Catalogue Management, Customer Management, Trade Account Approval, Pricing Engine

Category	Page Count	Includes
		Console, Promotions/Discounts, Delivery/Logistics Management, Returns/RMA, Reviews Moderation, Reporting & Analytics, Compliance Document Library

Total distinct templates: 71. Multiplied across real catalogue content (2,044 product detail pages + 21 category listing pages + N blog posts), the live sitemap runs into the low thousands of URLs — handled via ISR (Incremental Static Regeneration) in Next.js so this never becomes a build-time bottleneck.

07 — Landing Page Sections

The homepage ships with 16 distinct sections — exceeding the 12+ requested — each independently manageable from the CMS so merchandising can reorder, hide, or swap sections without a code deploy.

#	Section	What it shows
1	Hero Slider	7-slide rotating hero (already built for the corporate site; extended with shoppable CTAs)
2	Trending Now	Real-time best-selling SKUs, updated from order velocity
3	Recent Arrivals	Newest catalogue additions (e.g. the 7 newly-added categories)
4	Bundles & Combos	Curated multi-SKU bundles (e.g. “Patio Starter Kit”: pavers + jointing sand + sealer)
5	Clearance Sale	Discontinued colourways, overstock, and made-to-order cancellations at reduced price
6	Shop by Category	All 21 categories, grouped into the 4 product families
7	Shop by Sector	Commercial / Industrial / Institutional / Civil / Residential entry points
8	Colour-Matched Collections	Face + semi-face + paver groupings by colourway, the core differentiator
9	Best Sellers	Highest-rated and highest-volume SKUs
10	New to the Range	AAC, CSEB, Brick Slips, Braai Kits — the newest categories, explained
11	Customer Favourites	Ratings-driven carousel (4.5★+ only)
12	Project Inspiration Gallery	Real/representative imagery of finished work, linking back to the SKUs used
13	Delivery Coverage Map	Interactive map of the 7 served provinces and yard locations
14	Trust & Compliance Strip	SANS/SABS/NRCS/Agrément badges, payment logos, delivery partner logos
15	Newsletter & Trade Sign-Up	Email capture + trade-account application CTA
16	Recently Viewed	Personalised, session-based (returning visitors)

08 — Product Catalogue & Product Detail Pages

Every SKU in the Master Product Catalogue gets a dedicated, individually-addressable product page — not a shared variant selector — so each of the 2,044 SKUs is directly linkable, indexable, and reviewable on its own.

Feature	Detail
Dedicated product page	Every one of the 2,044 SKUs gets its own URL/page, generated from the Master Product Catalogue — not a shared configurator page
Image gallery	Multiple angles + zoom; colour-variant SKUs swap gallery images on swatch selection
Colour-matched cross-linking	Face brick ↔ semi-face brick ↔ matching paver, surfaced directly on the PDP (the core brand differentiator, made functional)
Technical specification table	Pulled straight from the catalogue's own columns: Applicable Standard, Size, Duty/Load Class, Sectors Served, Fulfilment Type
Downloadable spec sheet	Auto-generated PDF per SKU via the Python/FastAPI document service
Tiered pricing display	Retail / Trade / Volume prices shown live from the Pricing Engine; trade-tier price highlighted automatically for logged-in trade accounts
Coverage / quantity calculator	For pavers, slabs, cobbles: enter m ² → get required unit count using real per-category coverage rates
Stock & lead-time indicator	Stock Item vs Made-to-Order shown clearly, with the 15% MTO surcharge reflected in price before add-to-cart
Delivery estimate widget	Postcode entry on the PDP itself returns an ETA + fee preview before checkout
Ratings & reviews	Star summary, verified-purchase badge, photo reviews, helpfulness voting
Q&A section	Customer questions with staff/verified-owner answers, searchable
Frequently bought together	Bundle-style upsell (e.g. paver + jointing sand + sealer)
Related / recommended products	Powered by the Python/FastAPI recommendations service (Section 10)

Product Detail Page — Wireframe

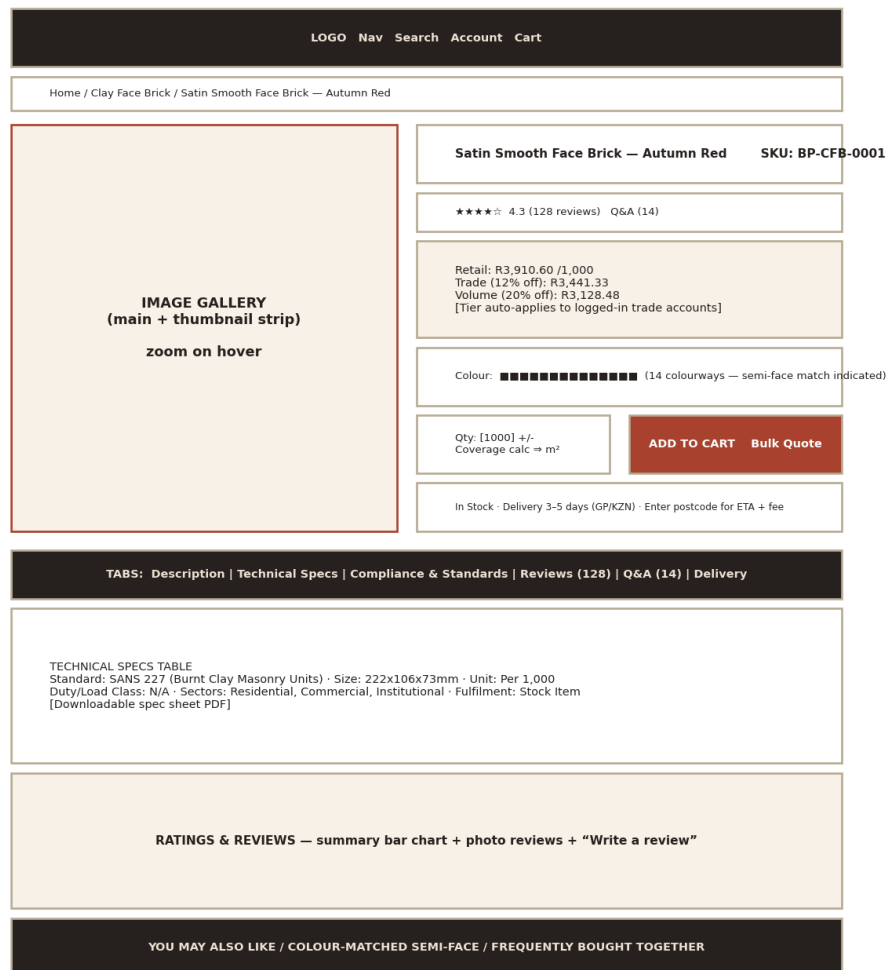


Figure 3 — Product Detail Page wireframe

09 — Search & Product Discovery

Search runs on Typesense (self-hosted on Render) or Meilisearch — both open-source, typo-tolerant, sub-50ms search engines well suited to a 2,044-document catalogue without the per-query cost of Algolia.

- Instant autocomplete/typeahead with product thumbnails, not just text suggestions
- Typo-tolerant full-text search across product name, category, colour, SKU and technical standard
- Faceted filtering: category, colour/finish (14), size, duty/load class, price range, fulfilment type, sectors served, applicable standard — all facets computed live from catalogue attributes
- Natural-language-ish queries handled via synonym mapping (e.g. “pavers for driveway” → Light Vehicular/Heavy-Duty duty class filter applied automatically)
- “No results” fallback surfaces the closest matching category and a bulk-quote CTA rather than a dead end
- Search index kept in sync via a BullMQ job triggered on any catalogue or pricing change

10 — Recommendations Engine

A dedicated Python/FastAPI microservice serves three recommendation surfaces from one underlying model set:

- Content-based: colour/category/attribute similarity — this is what powers the colour-matched face/semi-face/paver cross-linking, the platform's core differentiator
- Collaborative filtering: “customers who bought this also bought”, trained on order history as volume accumulates post-launch
- Merchandiser-curated bundles: manually-defined “frequently bought together” sets for common project combinations (e.g. paver + sand + sealer), editable from the CMS without a deploy
- Cold-start handling: for the first months of real order data, recommendations fall back to content-based similarity and merchandiser curation only — collaborative filtering activates once there's enough order volume to be statistically meaningful

11 — Ratings, Reviews & Q&A

- Star rating (1–5) + written review, gated to verified purchasers by default, with an “unverified” path for pre-purchase questions routed to Q&A instead
- Photo upload on reviews (moderated before publish)
- Helpful/Not helpful voting, sorted by helpfulness or recency
- Q&A: customer-to-customer and staff-answered; staff answers badged “Bricksplaza Team”
- Review moderation queue lives in the Business Desk (Section 17), with automated profanity/spam filtering before human review
- Aggregate rating and review count surfaced on category/search listing cards, not just the PDP

12 — Payment Methods & Financial Systems

15 payment methods ship at launch, exceeding the 12+ requested and matching the range built for Roofsteel plus the additional wallet and BNPL options a larger, more consumer-visible catalogue justifies.

Method	Type	Why it's included
PayFast	Card, EFT, wallet aggregator	SA's most established gateway; broad bank + card coverage
Ozow	Instant EFT	Real-time bank-to-bank, no card needed
SnapScan	QR wallet	High mobile adoption in SA retail
Zapper	QR wallet	Alternative QR wallet, wide merchant base
Yoco	Card processing	Also usable at physical yard trade counters (POS)
Payflex	BNPL (retail)	4-instalment buy-now-pay-later for consumer orders
PayJustNow	BNPL (retail)	3-instalment alternative BNPL, different underwriting panel
Mobicred	Consumer credit facility	Recurring SA online credit product
RCS	Store / retail credit card	Card widely held by SA retail credit customers
Peach Payments	Card / alt-payment aggregator	Redundancy/failover option alongside PayFast
Lulapay	B2B BNPL	Trade-account credit for contractors and builders (same as Roofsteel)
EFT / Bank Transfer	Manual proof-of-payment	Large trade/volume orders paid outside instant-gateway rails
Trade Account / Purchase Order	Net 7 / 14 / 30 credit terms	Tied directly to the Customer Tiering system and credit-vetting workflow
Apple Pay	Wallet (via card processor)	One-tap checkout on iOS/Safari
Google Pay	Wallet (via card processor)	One-tap checkout on Android/Chrome

Float, Accounts & Tiering

- Float (prepaid balance): trade customers can load a balance and draw it down against repeat orders — full ledger (top-ups, deductions, running balance) visible in both the customer account and the Business Desk
- Trade Accounts: credit-vetted, net 7/14/30 terms, credit limit enforced at checkout, approval workflow managed from the Business Desk (Section 17)
- Tiering: Retail (0% off) → Trade (12% off) → Volume (20% off), inherited directly from the Pricing Engine, with a Platinum tier reserved for top cumulative-spend trade accounts as a future lever, not hard-coded at launch
- All balances, credit limits and tier assignments are server-authoritative — never trusted from client state — and re-validated at checkout

13 — Checkout & Order Flow

A 5-step checkout — Cart → Delivery → Payment → Review → Confirmation — with the delivery fee calculated live the moment an address or postcode is entered (Section 14 covers the calculation itself).

Cart & Checkout — Wireframe

LOGO Secure Checkout [SSL Secured]

STEP INDICATOR: 1. Cart → 2. Delivery → 3. Payment → 4. Review → 5. Confirmation

CART ITEMS

- line item, qty, price, colour, remove
- line item...
- line item...

DELIVERY ADDRESS

- saved addresses / new address
- postcode → smart delivery fee calc

DELIVERY METHOD

- Standard (3-5 days)
- Express (1-2 days, select metros)
- Collect from yard

PAYMENT METHOD (12+)

- ☐ PayFast ☐ Ozow ☐ SnapScan ☐ Zapper
- ☐ Yoco ☐ Payflex ☐ PayjustNow ☐ Mobicred
- ☐ RCS ☐ Peach Payments ☐ EFT ☐ Trade Account/PO
- ☐ Apple Pay ☐ Google Pay

ORDER SUMMARY

Subtotal	R x,xxx
Delivery fee	R xxx
VAT (15%)	R xxx
Total	R x,xxx

Promo code [_____]

TRUST BADGES
SSL · PCI-DSS · SANS/SABS · Payment partner logos

PLACE ORDER

Figure 4 — Cart & Checkout wireframe

- Guest checkout supported, with an account-creation prompt at the confirmation step (order history is preserved if the guest later registers with the same email)
- Cart persists server-side for logged-in users (cross-device) and in local storage for guests
- Abandoned-cart email sequence (BullMQ scheduled job) for logged-in users who leave items unpurchased
- Order confirmation page and email both fire from a single server-side order-created event, so the two can never show conflicting information

14 — Delivery & Logistics

Delivery fees are calculated live at checkout — never a flat guess — using the same distance-banded logic already defined in the Pricing Engine's Delivery & Additional Charges sheet, extended here with real carrier integrations and tracking.

Band	Distance	Standard Time	Fee
Local	0–30 km	Same day / next business day	R950 flat (up to standard palletised load)
Regional	30–100 km	1–2 business days	R1,850 flat
Extended	100–250 km	2–4 business days	R3,200 flat
Long-distance	250 km+ (online-first provinces)	4–7 business days	Quoted live at checkout via carrier API
Collection	N/A	Same day (subject to stock)	Free — collect from nearest yard

Delivery Partners

Shortlisted against real capability to move heavy, bulky masonry loads nationally — not general parcel couriers:

Partner	Type	Capability	Role
DSV South Africa	National road freight & courier	1,600+ towns/cities nationally; palletised and heavy/oversized cargo; dedicated building-products logistics vertical; myDSV live tracking portal with API/EDI integration	Primary national carrier — Trade & Volume tier bulk orders, all 7 provinces
Faber Vervoer	Specialist brick/cement/concrete transport	Est. 1989; tri-axle trailers with hydraulic cranes; 10–35 ton capacity; client portal for booking, cost calculation and real-time load tracking	Specialist option for palletised brick/block loads, especially estate/complex deliveries needing crane offload
Panamax Bulk Carriers	Nationwide bulk commodity transport	Building materials, mining & industrial bulk freight; fleet tracking with live ETAs; load compliance/chain-of-responsibility management	Backup/overflow capacity for large civil and industrial orders
Besfleet (Besbpo Group)	Internal trucking division	Group-owned fleet — candidate internal option once route density justifies dedicated trucks, consistent with the approach taken for Roofsteel	Internal option to evaluate as Gauteng/KZN volumes scale

Smart Delivery Fee at Checkout

- Customer enters address/postcode → system resolves distance band from the nearest qualifying yard or approved supplier location
- Cart weight/pallet count (derived from each SKU's real unit-of-sale data) combines with the distance band to return a live fee — not a postcode-only estimate
- Crane/Hiab offload and cut-to-size surcharges (already defined in the Pricing Engine) apply automatically where the cart contains applicable SKUs
- For long-distance (250km+) online-first-province orders, checkout shows “Quote will be confirmed within 1 business day” rather than blocking the order

Tracking & Notifications

- Order status timeline: Processing → Dispatched → In Transit → Out for Delivery → Delivered, shown in the account dashboard
- Carrier webhook/API polling (myDSV API, Faber Vervoer client-portal API, Panamax fleet-tracking feed) normalises each partner's status into one internal schema
- Email + SMS notification on each status change; live map view where the carrier API exposes GPS data

Customer Account Dashboard — Wireframe

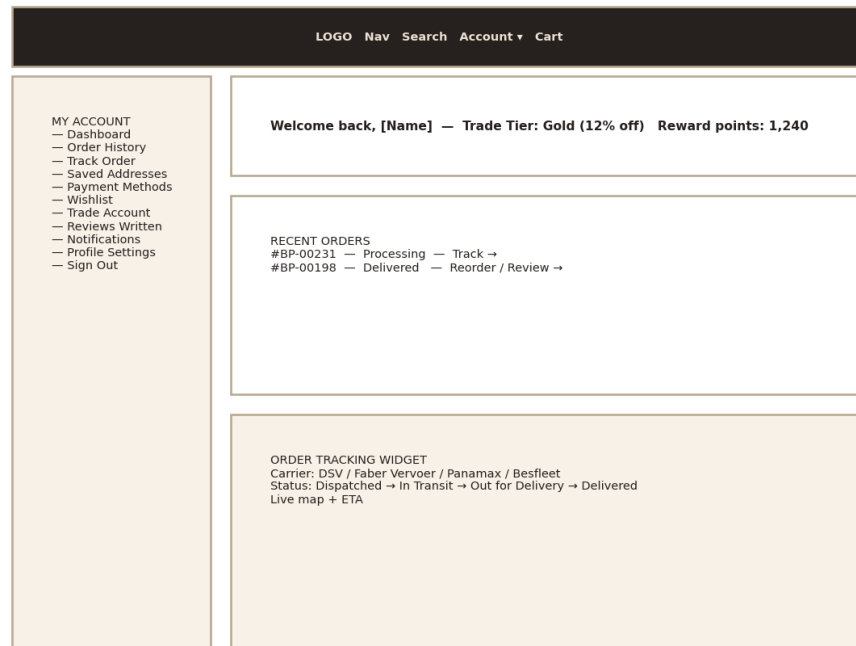


Figure 5 — Account dashboard with order tracking widget

15 — Trust & Credibility Placement

Credibility signals are placed where buying hesitation actually happens, not just dumped in the footer:

- Homepage trust strip (Section 7, position 14): SANS/SABS/NRCS/Agrément badges, payment logos, delivery partner logos — above the fold on mobile
- PDP: applicable SANS standard shown inline in the technical specs tab, not buried in a PDF
- Checkout: SSL/PCI-DSS badges and payment-partner logos placed directly beside the Place Order button, where cart-abandonment research consistently shows they matter most
- Footer: full compliance/association badge set + delivery partner logos, for the minority of visitors who scroll to verify
- Category pages for regulated categories (AAC, CSEB) show the Agrément SA certification note directly on the listing card, not just the PDP

16 — Content Management System (CMS)

A custom-built CMS (not a third-party headless CMS) so it can read/write directly against the same Postgres schema as the storefront — avoiding a second source of truth for product content, consistent with the approach taken on Roofsteel.

- Rich content editor for blog posts, category descriptions, and landing-page section copy
- Media library with automatic image optimisation/resizing on upload
- Promotion & banner manager — schedule homepage hero slides and clearance banners without a deploy
- SEO metadata manager — per-page title/description/OG-image overrides
- Draft/publish workflow with a preview link for content review before it goes live

17 — Business Desk (Admin Back-Office)

The operational nerve centre — order management, inventory, customer and trade-account administration, pricing control, and reporting, all role-gated by staff permission level.

Business Desk (Admin Back-Office) — Wireframe

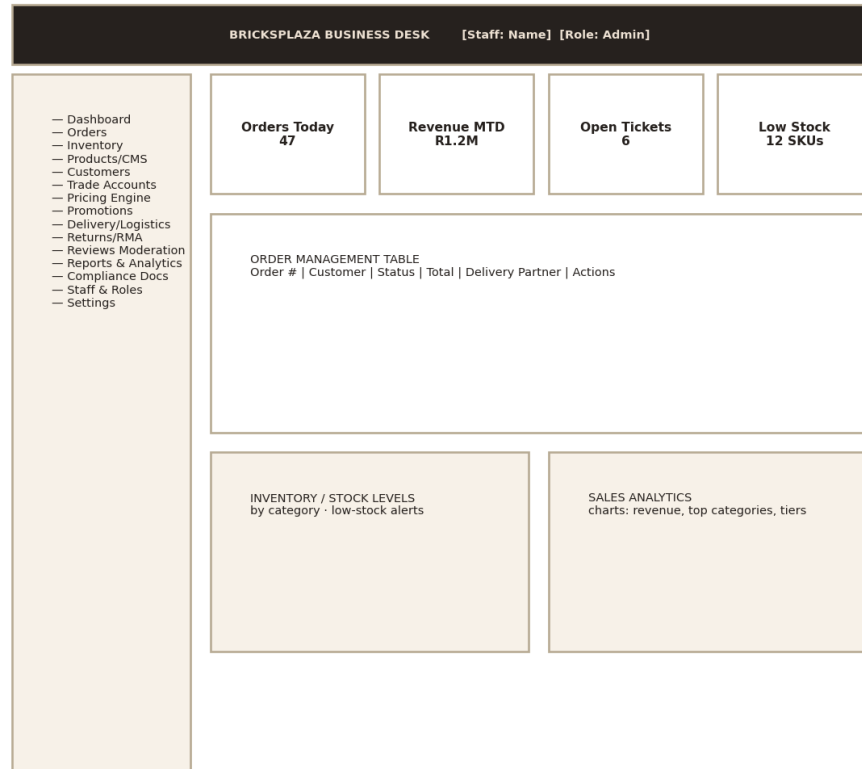


Figure 6 — Business Desk dashboard wireframe

- Order management: search/filter by status, customer, delivery partner; manual status override; refund/cancellation workflow
- Inventory management: stock levels by SKU, low-stock alerts, Stock Item/Made-to-Order flag management
- Trade account approval: credit-vetting workflow, credit limit assignment, tier upgrade/downgrade
- Pricing Engine console: direct read/write access to the Category Markup Bands, Multipliers and Customer Tiers already defined in the Pricing Engine v1.0 — changes here flow straight to the live storefront
- Reviews moderation queue, returns/RMA processing, and a compliance document library (SABS certificates, Agrément certificates, Letters of Authority) attached per supplier
- Reporting & analytics: revenue, top categories, tier mix, delivery-partner performance, all exportable

18 — Data Model Overview

Core entities (full schema detail is a build-phase artifact, not reproduced line-by-line here):

- User — auth identities (1:many across the 6 auth methods), profile, role
- Customer — tier, credit limit, float balance, saved addresses/payment methods
- Product — mirrors the Master Product Catalogue's 12 columns (SKU, Category, Product Type, Product Name, Colour/Finish, Size, Thickness, Unit of Sale, Applicable Standard, Duty/Load Class, Sectors Served, Fulfilment Type)
- PriceRule — mirrors the Pricing Engine's Category Markup Bands, Multipliers and Sundries Pricing sheets
- Order / OrderLine / Payment / Shipment — standard commerce entities, with Shipment carrying the carrier reference and tracking status
- Review / Question / Answer — linked to Product and verified-purchase Order
- ContentBlock / Promotion — CMS-managed homepage/landing content

19 — File Structure

```
bricksplaza-platform/
├── apps/
│   ├── web/
│   │   ├── app/
│   │   ├── components/
│   │   ├── lib/
│   │   └── public/
│   └── admin/
│       ├── app/
│       └── components/
├── services/
│   ├── api/
│   │   ├── src/modules/
│   │   └── src/common/
│   ├── recommendations/
│   ├── search-indexer/
│   └── doc-generator/
├── workers/
│   ├── email/
│   ├── delivery-sync/
│   └── search-reindex/
├── packages/
│   ├── ui/
│   ├── config/
│   └── types/
├── infra/
│   ├── terraform/
│   └── render.yaml
├── AGENTIC_RULES.md
├── docker-compose.yml
└── pnpm-workspace.yaml
```

Next.js 14 storefront (customer-facing)
App Router: (marketing), (shop), (account), (checkout)

API clients, auth helpers

Next.js Business Desk (role-gated)
(dashboard), (orders), (catalogue), (customers), (reports)

NestJS API gateway
auth, catalogue, pricing, orders, payments, delivery, reviews
guards, interceptors, pipes

Python/FastAPI

Python/FastAPI

Python/FastAPI (spec sheets, invoices)

BullMQ job processors (Node)

shared design-system components (web + admin)
shared eslint/tsconfig/tailwind config
shared TypeScript types (catalogue, order, user)

Render service definitions

local Postgres/Redis/Typesense

20 — LoC & File Estimates

Benchmarked against Roofsteel's real, shipped 35,521 LoC. Bricksplaza's build is materially larger in scope — 5 OAuth providers vs fewer, 15 payment methods vs 3, a full CMS and Business Desk, dedicated recommendation and search microservices, and a catalogue over 10× the size — so the estimate scales up accordingly.

Module	Estimated LoC	Estimated Files
apps/web (storefront)	18,000	280
apps/admin (Business Desk)	10,000	150
apps/api (NestJS gateway)	16,000	220
services/recommendations (FastAPI)	2,500	35
services/search-indexer (FastAPI)	2,000	30
services/doc-generator (FastAPI)	1,800	25
workers (BullMQ)	2,200	30
packages/ui (shared components)	5,000	90
packages/config, types, utils	2,000	40
infra (Terraform, CI/CD)	1,500	35
tests (unit, integration, e2e)	6,000	120

Total: ~69,000 LoC (range 65,000–72,000) across ~1,055 files. This is a working estimate for planning/resourcing purposes, not a hard ceiling — refine once Phase 1 scaffolding is real and measurable.

21 — Agentic Development Rules (Human-in-the-Loop)

This build uses HITL agentic development: AI agents (e.g. Claude Code) scaffold and implement against this specification; every rule below exists to keep a human decision-maker in control of anything that matters.

Rule	What it means in practice
AI proposes, humans approve	No agent merges its own pull request. Every change — schema, endpoint, UI, config — opens a PR for human review before merge, per Group-wide standing governance principle.
Scope discipline	Each agent task maps to one ticket/spec section. An agent that discovers unrelated issues files a new ticket rather than expanding its own diff.
No direct production access	Agents operate against local/staging environments only. Production deploys are a human-triggered promotion of a reviewed, tested build.
Migrations are gated	Database migrations are generated by agents but applied only after explicit human approval — never auto-applied on merge, even to staging.
Secrets never touched by agents	API keys, payment credentials and OAuth secrets are set directly by a human in Render's environment config; agents work against <code>.env.example</code> placeholders only.
Tests required before merge	Every PR includes unit tests for new logic and, for API endpoints, an integration test. CI must pass before human review begins.
Design fidelity check	UI PRs are checked against the wireframes in this spec (Section 8, 13, 14, 17) before merge — an agent flags any deliberate deviation for human sign-off rather than silently reinterpreting layout.
Payment & pricing changes get extra review	Any PR touching the payment gateways, pricing engine integration, or trade-account credit logic requires two human approvals, not one.
Traceable commits	Every agent commit message references the spec section or ticket it implements, so a human reviewer can trace a diff back to a requirement without re-reading the whole codebase.
Rollback plan required	Any PR that changes a data model or a payment/delivery integration states its rollback plan in the PR description before it can be merged.

22 — Phased Roadmap

Phase	Scope
Phase 1 — Foundation	Monorepo scaffold, auth (all 6 methods), catalogue import (2,044 SKUs), basic PDP/PLP, Postgres schema, Render environments (dev/staging/prod)

Phase	Scope
Phase 2 — Commerce Core	Cart, 5-step checkout, Pricing Engine integration, 5 initial payment gateways (PayFast, Ozow, Yoco, EFT, Trade Account), order management in Business Desk
Phase 3 — Discovery & Trust	Search (Typesense), faceted filtering, recommendations service v1 (content-based), ratings/reviews/Q&A, remaining 10 payment methods, credibility badge placement
Phase 4 — Delivery & Operations	Delivery partner API integrations (DSV, Faber Vervoer, Panamax), smart delivery fee calculator, tracking notifications, CMS content editor, full Business Desk reporting
Phase 5 — Launch Hardening	Load testing at catalogue scale, accessibility audit (WCAG 2.1 AA), POPIA/legal page sign-off, SEO technical audit, phased rollout: Gauteng/KZN first, then online-first nationwide

23 — Open Questions & Next Steps

- Confirm final choice between Typesense and Meilisearch once Render hosting costs for each are compared at expected query volume
- Confirm which of the 15 payment gateways sign contracts before Phase 2 vs which are added incrementally through Phase 3
- Confirm delivery-partner API access (myDSV API credentials, Faber Vervoer portal API terms) before Phase 4 integration work begins
- Decide whether the Business Desk ships as a fully separate Next.js app (as specified) or a role-gated section of the main app — separate app is recommended for security isolation, but adds a second deploy target on Render
- Legal review of the 9 legal-category pages (Section 6) before Phase 5 launch, particularly POPIA data-subject-rights language and trade-account credit terms

24 — Appendix: Scaffold Package Contents

A companion ZIP file accompanies this specification with starter scaffolding to accelerate Phase 1:

- /design — logo assets (SVG + PNG, all variants), Kiln palette as CSS variables and JSON tokens
- /content — homepage section copy stubs, legal page templates, product-page content schema
- /contacts — Besbpo Group division contact reference for the Group ecosystem footer/pages
- /codebase-structure — the monorepo folder tree (Section 19) with README stubs per module, package.json/pnpm-workspace.yaml scaffolding, docker-compose.yml for local Postgres/Redis/Typesense, and AGENTIC_RULES.md as a standalone reference file for the build agents